Name: ______________________

TOPIC TEST

Further functions (Advanced)

- Time allowed: 45 minutes
- Part A: 10 multiple-choice questions (10 marks)
- Part B: 8 free-response questions (40 marks)
- Total: 50 marks

Part A

10 multiple-choice questions
1 mark each: 10 marks
Circle the correct answer.

1 Which statement is true about an asymptote?

A It is a line that a curve will cross only once.

B It is the x-axis and any line parallel to it.

C It is the y-axis or any line parallel to it.

D It is a line towards which a curve approaches but never touches.

2 The frequency of a vibrating guitar string varies inversely as the length. A guitar string of length 0.7 m vibrates 3.7 times per second. What frequency would a string of 0.5 m length give in times per second?

A 2.627

B 5.254

C 1.314

D 3.5

3 Find the domain and range of $y = \frac{5}{x-1}$.

A Domain: $(-\infty, 0) \cup (0, \infty)$, range: $(-\infty, 0) \cup (0, \infty)$

B Domain: $(-\infty, 1) \cup (1, \infty)$, range: $(-\infty, 0) \cup (0, \infty)$

C Domain: $(-\infty, 6) \cup (6, \infty)$, range: $(-\infty, 0) \cup (0, \infty)$

D Domain: $(-\infty, 1) \cup (1, \infty)$, range: $(-\infty, \infty)$

4 Find the asymptotes of the graph of $y = \frac{3}{x+5}$.

A $x = 3, y = 0$

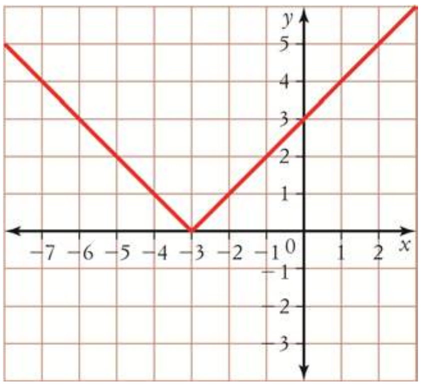
B $x = 0, y = 3$

C $x = -5, y = 0$

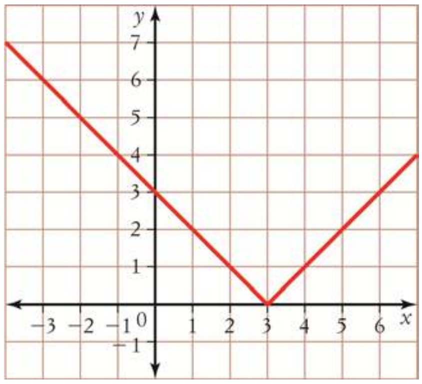
D No asymptotes

5 Which of these is the graph of $y = |x| + 3$?

A



B



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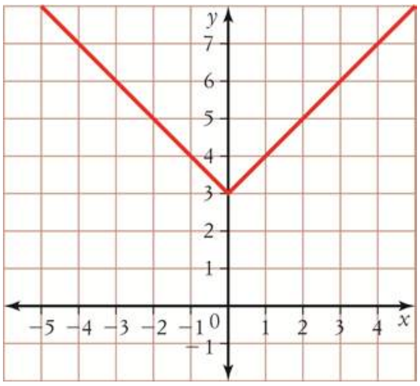
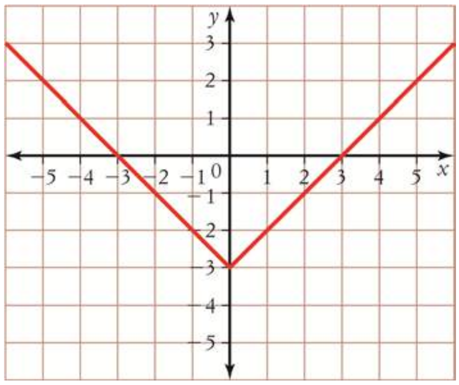


D



6 Find the x-intercepts and the y-intercepts of the graph of the function $y = 2|x - 8|$.

A $x = \pm 8$, $y = -16$ **B** $x = -8$, $y = 16$

C $x = 8$, $y = \pm 16$ **D** $x = 8$, $y = 16$

7 Find the centre O and the radius r of the circle with equation $x^2 + 4x + y^2 - 8y + 5 = 0$.

A $O(-2, 4)$, $r = \sqrt{15}$ **B** $O(2, -4)$, $r = \sqrt{15}$

C $O(-2, 4)$, $r = 15$ **D** $O(4, -2)$, $r = 15$

8 Which one of these functions has domain $[-5, 5]$ and range $[-5, 0]$?

A $y = \sqrt{5 - x}$ **B** $y = \sqrt{25 - x^2}$

C $y = \sqrt{x} - 5$ **D** $y = -\sqrt{25 - x^2}$

9 If $f(x) = x^3 + 3x - 1$ and $g(x) = x^2 - 1$, find the degree and the constant term of the polynomial $f(x)\,g(x)$.

A Degree 5, constant term 1

B Degree 5, constant term –1

C Degree 6, constant term –1

D Degree 6, constant term 1

10 For which pair of functions is $f(g(x)) = \sqrt{x^2 - 5}$?

A $f(x) = \sqrt{x} - 5$, $g(x) = x - 5$

B $f(x) = x^2$, $g(x) = \sqrt{x - 5}$

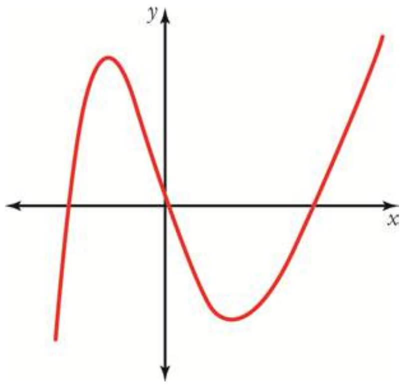
C $f(x) = x^2 - 5$, $g(x) = \sqrt{x}$

D $f(x) = \sqrt{x}$, $g(x) = x^2 - 5$

Part B

8 free-response questions
40 marks
Show your working where appropriate.

11 The graph of $y = f(x)$ is shown.



a Sketch the graph of $y = f(-x)$.

b Sketch the graph of $y = -f(-x)$.

[4 marks]

12 A shipping company makes this claim to its customers: 'The more you send, the less you pay per kilogram.' The shipping cost per kilogram ($\$C$) is inversely proportional to the mass (n kg) of the items shipped. When 5 kg of items are shipped, the company charges $7 per kg.

a Find the equation of C in terms of n.

b Find the cost per kg of shipping for 25 kg of items.

c How many kilograms are shipped if the cost per kg is $0.50?

d Sketch the graph of the cost per kilogram as a function of mass.

e Is it feasible for this shipping company to continue with this model? Discuss by giving examples and showing some calculations.

[6 marks]

13 Draw the graph of the function $y = x^3 - 1$ and $y = -f(x)$ on the same number plane.

[2 marks]

14 Find the domain and range of each function.

a $f(x) = |2x| - 2$

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b $f(x) = -|x - 5|$

____

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[2 marks]

15 Graph each function, showing clearly all intercepts.

a $y = |2x + 1| - 3$

b $y = 4 - |x|$

[4 marks]

16 Graph $(x-3)^2 + (y+2)^2 = 9$, clearly labelling all important points, and state the domain and range.

[3 marks]

17 Find the centre and radius of the circle with equation $x^2 + y^2 - 6y - 16 = 0$, then sketch the circle.

[3 marks]

18 Find, in expanded form, the equation of a circle with centre:

a (–3, 2) and radius 5

b (–2, 0) and radius 7

[4 marks]

19 $f(x) = 2x^3 - x^2 - 4x + 2$ and $g(x) = 2x - 1$. Find:

a $f(x) + g(x)$

b $f(x) - g(x)$

c $f(x)\,g(x)$

d $\dfrac{f(x)}{g(x)}$

[5 marks]

20 $f(x) = 2x - 4$ and $g(x) = x^2 - 4$.

a Find $\dfrac{f(x)}{g(x)}$ in its simplest form.

b Find the domain and range of $\dfrac{f(x)}{g(x)}$.

[4 marks]

21 Find $f(g(x))$ and $g(f(x))$, given that $f(x) = x^3$ and $g(x) = x + 1$, then find the domain and range of $f(g(x))$.

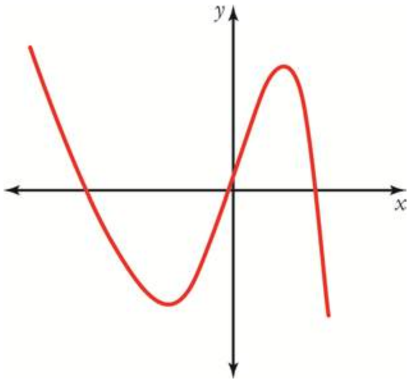
[3 marks]

This is the end of the test.
Use the rest of this page for extra working space.

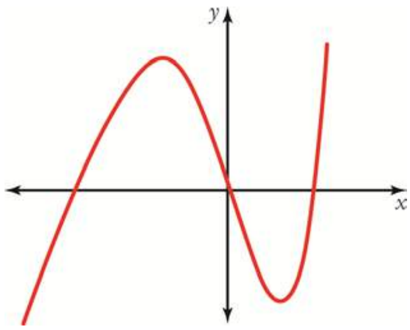
Answers

1 D	**2** B	**3** B	**4** C	**5** C
6 D	**7** A	**8** D	**9** A	**10** D

11 a $y = f(-x)$

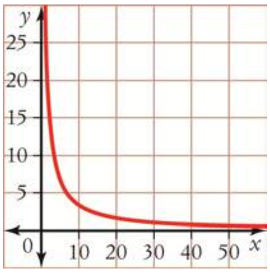


b $y = -f(-x)$



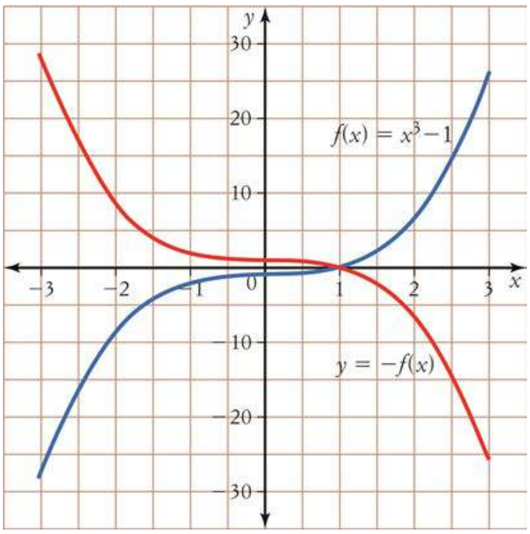
12 a $C = \dfrac{35}{n}$ **b** $C = \$1.4$ **c** $n = 70$ kg

d



e Not feasible, e.g. for 3500 kg the company would charge $0.01.

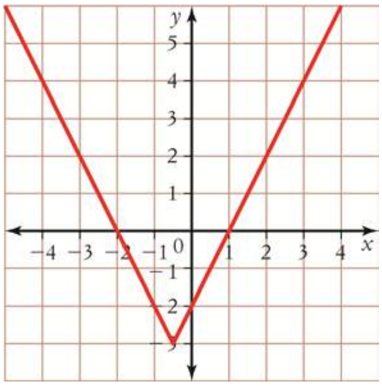
13



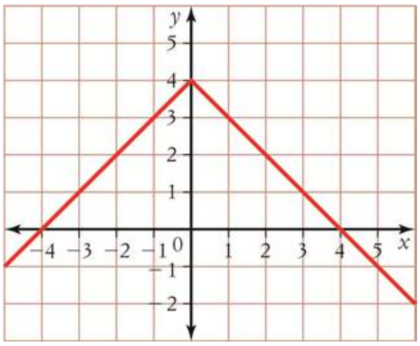
14 a Domain $(-\infty, \infty)$, range $[-2, \infty)$

b Domain $(-\infty, \infty)$, range $(-\infty, 0]$

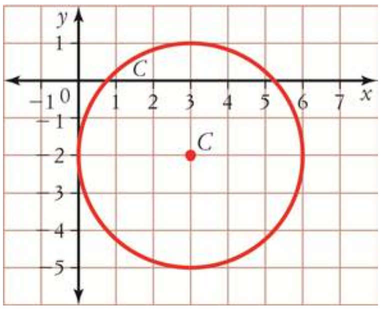
15 a



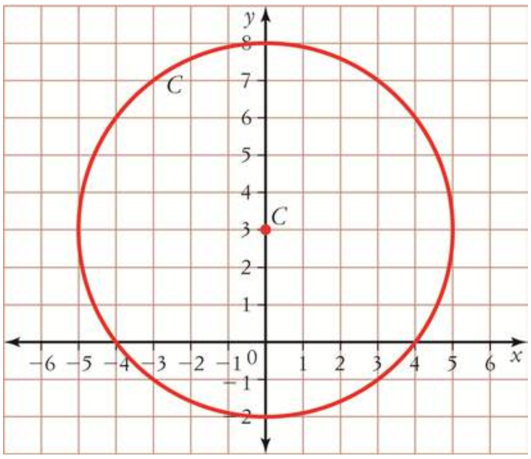
b



16 Domain [0, 6], range [–5, 1]



17 Centre (0, 3), radius 5



18 a $x^2 + 6x + y^2 - 4y - 12 = 0$ **b** $x^2 + 4x + y^2 - 45 = 0$

19 a $2x^3 - x^2 - 2x + 1$ **b** $2x^3 - x^2 - 6x + 3$

c $4x^4 - 4x^3 - 7x^2 + 8x - 2$ **d** $x^2 - 2$

20 a $\dfrac{2}{x+2}$

b Domain $(-\infty, -2) \cup (-2, 2) \cup (2, \infty)$, range $(-\infty, 0) \cup (0, \infty)$

21 $f(g(x)) = (x + 1)^3$, $g(f(x)) = x^3 + 1$; Domain $(-\infty, \infty)$, range $(-\infty, \infty)$